Universidad Nacional Abierta y a Distancia
Vicerrectoría Académica y de Investigación
Course: Advanced Machine Learning
Code: 2032238427

Learning Guide - Phase 4 – Time Series

1. Information of the Phase 4

Table 1. *Description.*

Aspect	Description
1. Type of activity	Independent
2. Evaluation moment	**Intermediate**
3. Managing unit	Escuela de Ciencias Básicas Tecnología e Ingeniería ECBTI
4. Score of the Choose an option	**100**
5. The activity starts on:	Wednesday, April 15, 2026
6. The activity ends on:	Tuesday, May 12, 2026
7. Student independent work hours	**20**

2. Detailed Description of the Learning Activity

Through this activity, the following learning outcome is expected to be achieved*:*

Learning Outcome 4: Build time series models for the prediction of time variables.

The activity consists of:

The student must develop a total of 4 exercises individually related with Time Series models. The Activities can be done using the Jupyter Notebook tool, which can be downloaded to your computers and work locally.

The exercises are based on the datasets corresponding to the chosen student:

Student 1: Monthly milk production

Student 2: Chemical concentrations readings

Student 3: Daily min temperatures

Student 4: Inflation consumer prices annual

Student 5: IBM common stock closing prices

You must submit a single document with the 6 exercises developed in the format **Appendix 1 – Template_Phase4**.

The following materials and resources are required for the development of this activity:

- Desktop or laptop computer. Internet connection.
- Python software installed on the computer or Google Colab account.
- Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and practice (3rd ed.). OTexts. (pp 1 - 90) https://otexts.com/fpp3/
- Kumar, N., Chauhan, R., & Dubey, G. (2021). Forecasting of Stock Price Using LSTM and Prophet Algorithm (Vol. 778). Springer Singapore. (pp 141 – 156) https://doi-org.bibliotecavirtual.unad.edu.co/10.1007/978-981-16-3067-5_12
- Appendix 1 – Template_Phase4.ipynb

Upload data and clean up

In a markdown cell make a brief description of the contents of the database, then use the Python pandas library to load the dataset corresponding to the selected Student and verify the presence of cells without values or outliers, as well as the type of data in each column and perform the respective cleaning. Adjust the dataset so that the column with time data is set as index.

Note: If there are more than two variables, apart from the time column, choose one of them for time analysis and justify the interest of your choice.

To develop this activity, follow these steps:

Step 1: Using a lineplot to study variable in the dataset, where on the horizontal axis is the time index. Then generate the columns 'Year', 'Month' and 'Day' and make a visualization of the different units of time using the subplot command, with which you can visualize the different graphs using different colors in them. With the previous visualizations, in a Markdown cell indicate whether or not there are trends, seasonality or cyclical changes in the data, justify your response using new graphs if necessary.

Step 2: Analyze if the dataset meets the three conditions for stationarity: Zero trend, constant variance, and constant autocorrelation by means of graphs. Apply the Augmented Dickey-Fuller test and interpret the results. Make an autocorrelation plot. If it is not stationary apply First-order differencing and Second-order differencing (If necessary), if the series is seasonal apply Seasonal differencing. Split the stationary series into two Train/Test groups, taking 70% for training. Graphically visualize split data in Tran/Test.

Step 3: With the original Time Series Data, with resampling if it was necessary, optimize the ARIMA parameter with the value AIC. Building an optimized model ARIMA(p,d,q) and plotting the model with the original data in the same graphic.

Step 4: With the same Train/Test split of the step 2, build a model by Prophet with the parameters given by default and compare, both graphically and with their respective metrics, with the previous models. Optimize PROPHET model parameters and compare.

Step 5: Using the same Train/Test split from step 2, build an LSTM neural network model to forecast the selected time series. Compare the results, both graphically and using the corresponding evaluation metrics, with the previous models.

Then, optimize the main LSTM parameters (number of neurons, sequence length, and epochs) and compare the performance.

3. Guidelines for Developing and Submitting Learning Evidence

Learning evidence refers to the actions, products, or observable processes that are done or delivered to demonstrate acquired capabilities, skills, aptitudes, and attitudes. These serve to allow the teacher to assess and evaluate student performance effectively.

The evidence to be developed individually are:

- The development of the four exercises, which you will have to perform in documents
- with ipynb extension, as requested in the delivery conditions.

For developing and submitting evidence, please consider the following guidelines:

1. All group members must participate with their contributions in the development of the activity.

2. Each group should designate one member to be responsible for submitting the requested product in the environment specified by the teacher.

3. Before submitting the requested product, the group should check that it complies with all the requirements indicated in this activity guide.

4. Only group members who have actively contributed during the allocated time for the activity should be included as authors of the requested product.

Please note that all individual or group written products must comply with "spelling and mechanics" standards and the submission conditions defined in this activity guide.

Regarding the use of references, consider that the product of this activity must comply with the standards of Choose an option.

In any case, comply with the referencing standards and avoid academic plagiarism. To do so, you can support yourself by reviewing your written products using the Turnitin tool available in the virtual campus.

4. Academic Situations

Consider that in Agreement 029 of December 13, 2013, Article 99, the following actions are considered as offenses against academic order, among others: item e) "Plagiarism, that is, presenting as one's own the entirety or part of a work, paper, document, or invention created by another person. It also includes the use of false citations or references, or proposing citations where there is no match between the citation and the reference," and item f) "Reproducing or copying, for profit, educational materials or results of research products that have intellectual property rights reserved for the University."

The academic sanctions that the student will face are as follows:

a) In cases of proven academic fraud in the respective academic work or evaluation, the grade imposed will be zero points without prejudice to the corresponding disciplinary sanction.

b) In cases related to proven plagiarism in academic work of any nature, the grade imposed will be zero points, without prejudice to the corresponding disciplinary sanction.